



SARAVAN PRASANNA

AI ENGINEER

Fresher 2nd year: Open to intern

Profile

Innovative AI Engineer experienced in the end-to-end development of autonomous systems and IoT-integrated AI. Strong background in C++, Python, and Three.js, with a proven track record of leading technical teams to deliver automated regulatory systems and social robotics. Committed to leveraging cutting-edge AI frameworks to solve complex industrial challenges and optimize user engagement.

Education History

➤ **Bachelor in Technology**
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE 2025 - 2029
V S B College Of Engineering
Technical Campus

➤ **Higher Secondary Schooling**
2024 - 2025
Sri Gopalnaidu Hr Sec School

Languages

- English
- Tamil
- Malayalam

Contact Info

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Skills

- Problem Solving
- Machine Learning
- Project Management
- Video Editing
- Website Developing
- Team Management

Courses completed

➤ **Front End Development - HTML | Great Learning Academy (May 2024)** Completed online coursework in web development fundamentals, mastering semantic HTML structure, core markup techniques, and web page layout design. Acquired practical knowledge in building clean, well-structured web elements as a foundation for modern front-end user interfaces.

➤ **Basics of Python | Infosys Springboard** Successfully completed coursework through Infosys Springboard focused on fundamental Python programming concepts. Gained core practical knowledge in writing syntax, utilizing essential data types, handling basic algorithms, and building foundational problem-solving skills for software development.

➤ **Course on Machine Learning (ML) | Nxtsync** Completed a comprehensive three-month Machine Learning program conducted by Nxtsync (AICTE Approved & ISO 9001 Certified). Gained hands-on experience in core ML concepts, data analysis techniques, predictive modeling, and practical algorithm implementation for data-driven applications.

Projects

AI-Powered Voice Assistant Robot (ESP32-Based Embedded AI System)

- Designed and developed a standalone, hands-free AI voice assistant using the ESP32 microcontroller, integrating embedded audio hardware with cloud-based artificial intelligence to enable real-time conversational interaction. Engineered the complete signal pipeline — capturing voice input via an I2S MEMS microphone, streaming audio over WiFi to a Python Flask backend for speech-to-text conversion, and generating context-aware responses through a large language model (Meta Llama 3.3 70B via Groq API), augmented with real-time web search (Tavily API) to overcome static AI knowledge cutoffs. Implemented text-to-speech synthesis and streamed audio playback through an I2S Class-D amplifier, while designing a reactive OLED display interface with custom-animated facial expressions synchronized to conversation states (listening, processing, and speaking). Independently diagnosed and resolved several low-level embedded systems challenges, including dynamic memory management for audio buffering under ESP32 RAM constraints, I2S peripheral clock conflicts between simultaneous audio input/output channels, and display-induced signal interference during real-time audio streaming. Built entirely using free-tier cloud services and open-source tools, demonstrating practical skills in embedded C++ programming, RESTful API integration, digital signal handling, and full-stack IoT system design.

LearnPath — Adaptive Learning Platform

- LearnPath is a full-stack, adaptive e-learning platform I designed and built to solve a common problem with free online learning resources: unstructured content, no feedback loops, and disconnected practice environments. The platform begins with a personalized onboarding flow that captures each learner's current skill stage, subject interests, and goals, and uses that profile to drive every other feature — a YouTube-powered video feed automatically organized into Beginner, Intermediate, and Expert tiers per topic, a weekly auto-generated assessment that rotates question sets and grades submissions instantly, and an integrated multi-language online compiler (supporting Python, C, C++, Java, and JavaScript) that lets learners immediately test what they've just studied against custom test cases with live pass/fail feedback. I built the backend using Python and Flask with a SQLite database, exposing a REST API consumed by a responsive HTML/CSS/JavaScript frontend, and implemented secure per-user authentication with hashed passwords and fully isolated user data.

Projects

- A live analytics dashboard, built with Chart.js, consolidates each learner's video-watch history, quiz score trends, and study streaks, along with a stage-advancement suggestion engine based on recent performance. To ensure reliability during demos and real-world use, I integrated the YouTube Data API v3 and the Judge0 code-execution API with automatic offline fallbacks, so the platform remains fully functional even without external API access. The project also includes a curated directory of free courses from Coursera, Udemy, Infosys Springboard, and NPTEL, and a built-in autosaving notepad with one-click download support, rounding out a complete, self-contained learning environment built entirely from scratch.

Smart Attendance and Emotion Analytics System | Python, OpenCV, Flask, SQLite, CNN, JavaScript |

- I designed and developed a full-stack, browser-based Smart Attendance and Emotion Analytics System that automates student attendance marking using real-time face recognition through a standard webcam, while simultaneously detecting and logging the emotional state of each recognized individual. The entire system operates on local hardware without any GPU, cloud subscription, or internet connection, making it practically deployable in any educational institution with just a laptop and a webcam.
- On the backend, I built a Python and Flask server that integrates three computer vision components working in parallel: an OpenCV Haar cascade classifier for real-time face detection, an LBPH (Local Binary Pattern Histograms) recognizer for face identification, and a lightweight convolutional neural network trained on the FER2013 dataset for seven-class facial emotion recognition covering Happy, Sad, Angry, Disgust, Fear, Neutral, and Surprise. Every detected face region is preprocessed using CLAHE (Contrast Limited Adaptive Histogram Equalization) followed by histogram equalization before being passed to both models, which ensures consistent and robust recognition performance under varying indoor lighting conditions. To eliminate false positives, I implemented a five-frame consecutive recognition filter that requires the same person to be identified in five successive frames before any attendance event is written to the database — this completely eliminated all false-positive attendance entries across all test sessions.

Projects

- All data is stored locally in a SQLite database with a clean normalized schema consisting of three tables — `persons`, `attendance_log`, and `emotion_log` — designed to support efficient date-range filtering and per-student aggregation without requiring any external database server. On the frontend, I built a five-module single-page web application using HTML, CSS, JavaScript, and Chart.js that communicates with the Flask backend through REST API calls. The Dashboard module displays live KPI cards showing total registered students, present and absent counts, attendance rate, and the dominant mood of the day, along with a seven-day attendance bar chart and an emotion distribution donut chart. The Live Attendance module streams the webcam feed in real time with bounding boxes, recognition labels, and emotion overlays drawn on detected faces, and a side panel logs every recognition event with the person's name, roll number, timestamp, and detected emotion. The Register Person module guides the operator through a five-sample webcam face enrollment process with numbered progress indicators, and the backend performs incremental LBPH retraining on submission so the new person is immediately recognizable without restarting the system. The Attendance Records module provides a fully filterable table of attendance history by student and date range with On-time and Late status badges and a one-click CSV export for integration with institutional management systems. The Emotion Analytics module displays a per-student emotion distribution donut chart, a heuristic wellbeing signal bar chart that aggregates negative emotion counts per person, and a detailed raw emotion log table with confidence scores for each reading.
- I evaluated the system across 1,350 test frames recorded over five separate sessions on different days with three enrolled subjects, introducing natural variation in lighting, pose, and appearance. The face recognition module achieved a macro-average precision of 94.73%, recall of 93.26%, and an F1-score of 93.99%. The emotion classification CNN achieved a weighted F1-score of 87.4% across all seven emotion classes, with Neutral and Happy performing best at 93.3% and 90.4% respectively. The end-to-end pipeline ran at approximately 20.5 frames per second on an Intel Core i5 11th Generation CPU with no GPU acceleration. I also authored a complete IEEE-format research paper documenting the system architecture, the CLAHE preprocessing methodology, the experimental setup, and the full precision, recall, and F1-score evaluation results for both the face recognition and emotion classification components.

Projects

WEBSITE FOR THE EDUCATIONAL PURPOSE

- Engineered a high-performance educational platform using a modern frontend stack (HTML5, CSS3, JavaScript), featuring a minimalist Glassmorphism-inspired UI. Implemented responsive design principles to ensure a seamless learning experience across all devices, prioritizing intuitive navigation and clean information architecture.

Achievements and Awards



Ideathon Winner | Inter-Collegiate Fest

Awarded First Place for conceptualizing and presenting an innovative solution during the first-year inter-collegiate Ideathon. Demonstrated strong analytical thinking, problem-solving skills, and the ability to pitch complex technical ideas to a panel of judges.



3rd Place: Code Debugging | Inter-Collegiate Fest

Secured Third Place in a competitive debugging challenge. Successfully identified and resolved complex logical, syntactical, and runtime errors across multiple programming languages under strict time constraints, showcasing deep technical comprehension and attention to detail.



3rd Place: Web Design | GIT Institution, Kottayam

Earned Third Place in a regional Web Design competition hosted by GIT Institution. Developed a responsive and user-centric website interface, balancing aesthetic appeal with functional UI/UX principles and modern front-end technologies.



Project Expo Participant | Inter-Collegiate Event

Selected to showcase a technical prototype during the first-year Project Expo. Engaged with industry experts and peers to demonstrate the project's functionality, hardware/software integration, and potential real-world applications.



Codeathon Participant | Inter-Collegiate Event

Competed in an intensive, fast-paced Codeathon, collaborating with a team to build functional software solutions within a limited timeframe. This experience refined my rapid prototyping skills and my ability to write clean, efficient code in high-pressure environments.



Innovation Catalyst – Intra College Project Contest

Participated in the intra-college project contest organized by the Electronics Club, presenting an innovative project recognized for outstanding technical ingenuity and execution. Demonstrated practical application of engineering and Artificial Intelligence principles, effective teamwork, and strong technical communication skills.



Sparkathon-6hrs Hackathon

Actively participated in "SPARKATHON," an intensive 6-hour non-stop hackathon, focused on developing innovative Internet of Things (IoT) solutions. Collaborated under tight deadlines to design, prototype, and present a practical IoT-driven concept, demonstrating strong real-time problem-solving, rapid prototyping, hardware-software integration, and effective teamwork.